

Browning Automatic Rifle M1918A2

PAGES FROM THE ORIGINAL MANUAL

FM 23-15, Basic Field Manual – Browning Automatic Rifle,
Caliber .30, M1918A2 (War Department, 1943)

WHAT'S INSIDE

Chapter 1, Mechanical Training: description, taking it apart and putting it back together, cleaning, how it works, operation, stoppages, spare parts and ammunition.

54 pages from the original follow: the manual's cover, then the section about this weapon. Original US Government publication (public domain). Scans courtesy of the Internet Archive and Wikimedia Commons.

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FM 23-15 AR DEPARTMENT

BASIC FIELD MANUAL

BROWNING AUTOMATIC RIFLE

CALIBER .30, M1918A2

30 June 1943

U.S. Army Military History Institute

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BASIC FIELD MANUAL

BROWNING AUTOMATIC RIFLE, CALIBER .30, M1918A2

This manual supersedes FM 23-15, 27 Aug. 1940; including C 1, 15 July 1941; C 2, 15 Nov. 1941; C 3, 10 July 1942; C 4, 5 Nov. 1942; and C 5, 18 Dec. 1942.

Attention is directed to FM 21-7 for details as to how appropriate training films and film strips are intended to be used and how they are made available for use during training with the Browning automatic rifle, caliber .30, M1918A2.

CHAPTER 1

MECHANICAL TRAINING

	Paragraphs
SECTION I. General.....	1-3
II. Disassembly and assembly.....	4-12
III. Care and cleaning.....	13-16
IV. Functioning.....	17-28
V. Operation.....	29-37
VI. Stoppages and immediate action.....	38-41
VII. Spare parts and accessories.....	42-43
VIII. Ammunition.....	44-51

SECTION I

GENERAL

■ 1. OBJECT.—This chapter is designed to give the soldier training that will insure his ability to maintain the rifle and keep it in operation.

■ 2. DESCRIPTION.—The Browning automatic rifle, caliber .30, M1918A2, is an air-cooled, gas-operated, magazine-fed shoulder weapon with bipod (fig. 1). It weighs approximately 20 pounds with sling. The ammunition is loaded in magazines of 20 rounds. The weight of the magazine when empty is 7 ounces; when filled, 1 pound 7 ounces.

■ 3. FIRE POWER.—The Browning automatic rifle, caliber .30, M1918A2, is not mechanically capable of semiautomatic fire, though it can be fired single shot by proper trigger manipu-

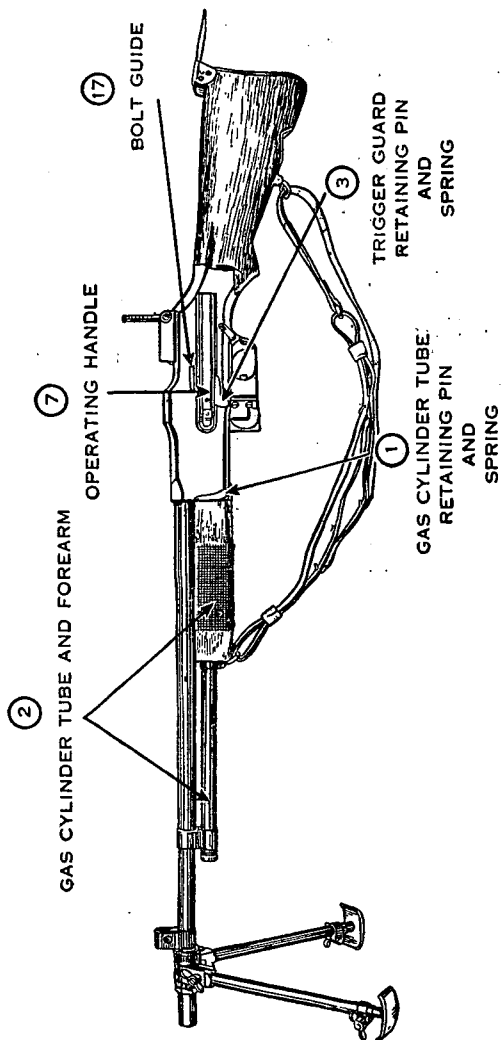


FIGURE 1.—Browning automatic rifle, caliber .30, M1918A2.

lation. There are two cyclic rates of full automatic fire, normal and slow, which may be selected by the firer. The normal cyclic rate is approximately 550 rounds per minute; the slow cyclic rate is approximately 350 rounds per minute. The most effective rate of fire for this weapon is from 120 to 150 rounds per minute. The sustained rate, however, is from 40 to 60 rounds per minute.

SECTION II

DISASSEMBLY AND ASSEMBLY

■ 4. TRAINING.—Training in disassembly and assembly will be taken up as soon as practicable after the soldier receives his rifle. In any case, it will be completed before any firing is done with the rifle by the individual. Instruction in the care and cleaning of the automatic rifle will also be covered.

■ 5. ORGANIZATION.—In the company or platoon, all enlisted men of the automatic rifle teams are combined in one or more groups, for instruction, under selected officer and non-commissioned officer instructors. Rifle squad leaders or assistant squad leaders supervise the work of their respective automatic rifle teams.

■ 6. CARE TO BE EXERCISED.—*a.* The rifle can normally be disassembled and assembled without applying undue force. When undue resistance is met, force will be applied cautiously.

b. The rifle will not be disassembled or assembled against time. Instruction, blindfolded, may be given to men who have passed their tests in mechanical training. In all work in disassembling the rifle, the men will be taught to lay out the parts on a smooth, clean surface from left to right, in the same sequence as they are disassembled from the rifle. The trigger mechanism will not be disassembled or assembled blindfolded.

■ 7. NOMENCLATURE.—The names of the parts to which reference is made in mechanical training are readily learned as this training progresses. Instructors will therefore take care to name the parts clearly and correctly in their work. A sufficient knowledge of the nomenclature of the rifle is

gained by the soldier during the instruction in mechanical training.

■ **8. DISASSEMBLING.**—*a. General.*—Authorized disassembly by the soldier is limited to that required for proper care and maintenance of the rifle. Further disassembly will be done under the supervision of an officer or ordnance personnel. The individual soldier is prohibited from disassembling the—

- (1) Forearm group.
- (2) Barrel group.
- (3) Butt stock and buffer group.
- (4) Rear sight group.
- (5) Receiver group.

b. Sequence.—The steps in the disassembly of the rifle authorized to be performed by the individual soldier without supervision are performed in the following sequence:

- (1) *Operating group.*—(a) Cock the rifle and remove—
 1. Gas cylinder tube retaining pin and spring.
 2. Gas cylinder tube and forearm.
- (b) Let the slide forward and remove—
 1. Trigger guard retaining pin and spring.
 2. Trigger guard.
 3. Recoil spring guide.
 4. Recoil spring.
 5. Operating handle.
 6. Hammer pin.
 7. Hammer.
 8. Slide.
 9. Bolt assembly.
 10. Firing pin.
 11. Bolt link pin.
 12. Bolt link.
 13. Extractor.
 14. Extractor spring.
- (2) *Trigger group.*—Remove—
 - (a) Ejector.
 - (b) Magazine catch spring.
 - (c) Magazine catch pin.
 - (d) Magazine catch.

- (e) Magazine release.
- (f) Sear spring.
- (g) Trigger pin.
- (h) Trigger.
- (i) Connector.
- (j) Sear pin.
- (k) Sear.
- (l) Sear release stop lever.
- (m) Sear carrier.
- (n) Counterrecoil spring.
- (o) Change lever.
- (p) Change lever spring.
- (3) *Bipod group*.—Remove—
 - (a) Flash hider.
 - (b) Two bipod leg clamp wing screws.
 - (c) Right and left bipod leg assemblies.
 - (d) Two bipod clamp leg guides.
 - (e) Two bipod body wing screws.
 - (f) Right and left bipod leg tubes.
 - (g) Bipod spring-lock washer.

c. *Method*.—The following detailed explanation of the method of disassembling the automatic rifle is furnished as an aid to instructors:

(1) *Operating group* (figs. 1 and 2).—Lay rifle on a table, barrel down, pointing to the left. Cock the rifle. Turn gas cylinder tube retaining pin spring (1) 180° in a clockwise direction and pull out gas cylinder tube retaining pin and spring (1). Remove gas cylinder tube and forearm (2). Let slide (10) forward *easily* by pressing trigger with thumb of the right hand and at the same time grasping the slide with the left hand so that the middle and index fingers are astride the gas piston. Turn trigger guard retaining pin spring (3) 90° in a clockwise direction and pull out the pin and spring (3). Lift out trigger group (4). Remove recoil spring guide (5) by pressing right index finger on checkered surface of its head and turning it until ends are clear of the retaining shoulders. Remove recoil spring (6). Line up hammer pin holes in receiver and operating handle by inserting point of recoil spring guide in hole in operating handle with right hand, press against hammer pin (8) until it clears the

operating handle by jutting through its hole in the receiver as the operating handle is pushed slowly to the rear with the left hand. Remove operating handle (7) by pulling straight to the rear. Remove hammer pin (8). Push hammer (9) forward out of its seat in the slide (10) and lift out of the receiver. Push the bolt link (13) to the rear and down. Being careful to avoid striking the gas piston or rings against the gas cylinder tube bracket, remove slide (10) by pulling it forward out of the receiver. Pry bolt guide (17) outward with the rim of a cartridge. Lift out bolt assembly (14) by pulling it slowly to the rear end of the receiver and raising it with the right thumb and forefinger. (As an alternate method of removing the bolt, move bolt assembly all the way to the rear end of receiver, bolt link up and forward. Then, grasp front end of bolt between left forefinger and thumb, rotate bolt assembly gently to clear bolt guide and bolt support on sides of receiver, and lift bolt assembly out of receiver.) Pull out firing pin (11) from its aperture in the bolt. Remove bolt link pin (12). Remove bolt link (13). Remove extractor (15) by pressing the point of a dummy cartridge against claw and exerting pressure outward and to the front. Remove extractor spring (16).

(2) *Trigger group* (fig. 3).—(a) Depress ejector lock with the point of a dummy cartridge. Hold thumb in front of magazine catch spring to prevent it from flying out, and then slide ejector (18) out of its seat. Remove magazine catch spring (19). Remove magazine catch pin (20), lift out magazine catch (21) and magazine release (22).

(b) Holding trigger guard so that its rear is away from the body, insert trigger guard retaining pin spring under the *three leaves* of the sear spring *above the connector stop*. Pry up so that center leaf trips above front end of sear. Rotating the trigger guard retaining pin forward while pressing firmly down and forward on sear spring with the left thumb, unseat the sear spring and pull it out to the front. Remove the trigger pin (24). The trigger pin must always be removed before the sear pin in order that tension of the counterrecoil spring will always be on the sear pin. Remove trigger (25) and connector (26) through top of the trigger guard. Push out sear pin (27) with the recoil

spring guide. Remove sear (28) and sear release stop lever (29). *Noting the manner in which these parts are fitted together prior to separating them*, separate the sear and sear release stop lever. Pry up on the sear carrier (30) by

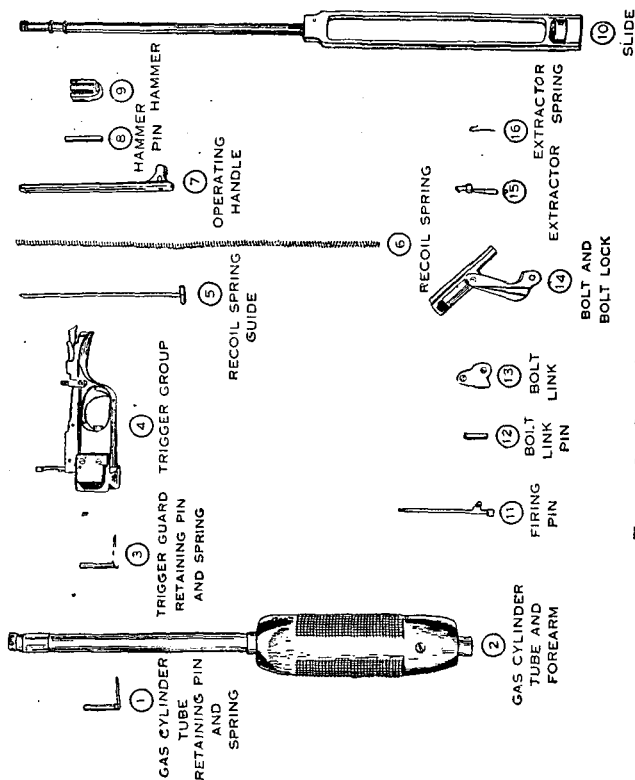


FIGURE 2.—Operating group.

inserting the point of the recoil spring guide in either of the sear pin holes in the sear carrier. Lift out the sear carrier and counterrecoil spring (31). Remove the change lever (32) by prying up on the forward end of the change lever spring with the rounded end of the sear spring held in the

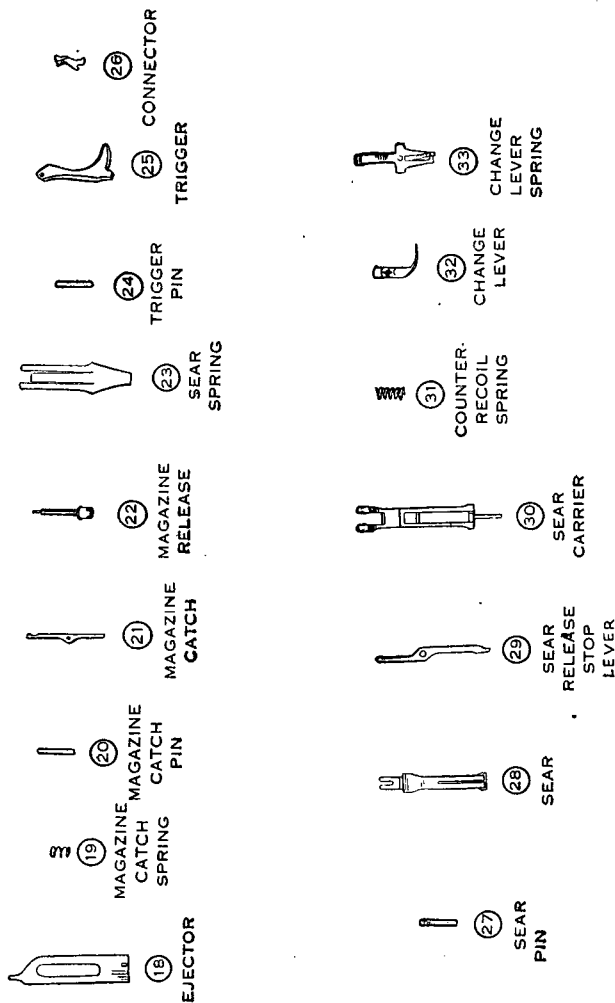


FIGURE 3.—Trigger group.

right hand, withdrawing the change lever to the left with the left hand. Being careful not to bend or break the sear release stop lever spring which is riveted to the change lever spring (33), remove the change lever spring by prying it to the rear with the rounded end of the sear spring so as to unseat the shoulders of the change lever spring from their slots in the trigger guard.

(3) *Bipod group* (fig. 4).—The disassembling of the bipod group generally is not performed as a part of mechanical training. However, it may become necessary to disassemble this group to replace damaged or broken parts, and the following procedure is prescribed: Unscrew the flash hider (34). Remove the bipod assembly. Unscrew the two bipod leg clamp wing screws (35). Remove right and left bipod legs (36). Remove the two bipod clamp leg guides (37). Unscrew the two bipod body wing screws (38). Remove right and left bipod leg tubes (39) from bipod body (40). Remove bipod spring-lock washer (41).

■ 9. ASSEMBLING.—The rifle and its component parts are assembled in the reverse order of their disassembly as given in paragraph 8c. The following detailed explanation of the method of assembling the rifle is furnished as an aid to instructors:

a. *Bipod group* (fig. 4).—Replace bipod spring-lock washer (41). Replace the right and left bipod leg tubes (39) on the bipod body (40). Screw the two bipod wing screws (38) into the bipod body. Replace the two bipod clamp leg guides (37) in their slots on the right and left bipod legs. Replace bipod legs (36) in the bipod leg tubes (39). Aline U-shaped opening of the bipod clamp leg guides with the holes in the bipod leg clamps, and replace the bipod leg clamp wing screws (35). Replace bipod assembly on the rifle. Screw flash hider (34) on to the rifle.

b. *Trigger group* (fig. 3).—(1) Replace the change lever spring (33) with sear release stop lever spring up and to the rear, by pressing down and forward on the rivet with the thumb of the right hand and engaging the ears of the change lever spring in their slots in the rear of the trigger guard. Then, *being careful not to bend or break the sear re-*

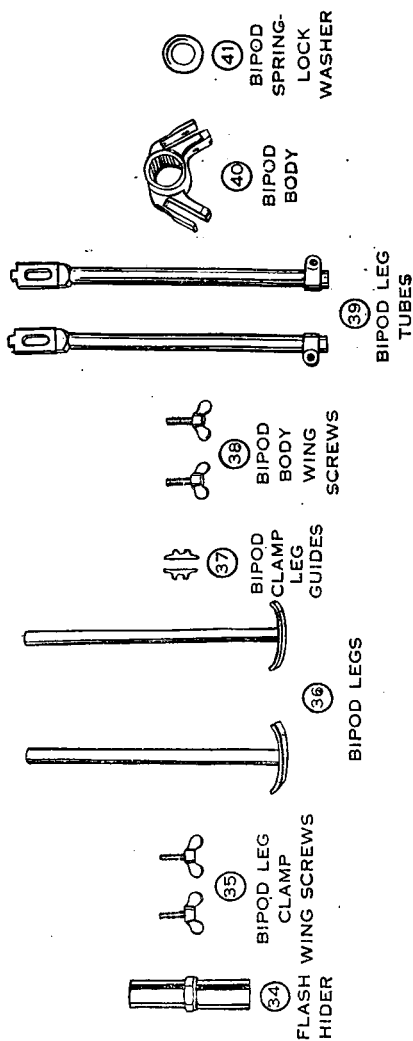


FIGURE 4.—Bipod group.

lease stop lever spring, continue the pressure with the right thumb, using rounded end of the sear spring as a ramp, and seat the rear end of the change lever spring in its slot. Replace change lever (32) by inserting it in its hole in the trigger guard with the left hand while with the right hand prying up on the forward end of the change lever spring with the rounded end of the sear spring. Replace counterrecoil spring (31) on counterrecoil spring guide (front end of sear carrier). Insert counterrecoil spring guide into its seat in the trigger guard. Brace forward end of trigger guard against an immovable object, and with the thumbs pressing on the rear of the sear carrier, push sear carrier (30) forward until its rear end is held by the ears of the change lever spring. Replace the sear release stop lever (29) in its proper seat in the sear (28) by holding the sear in the left hand with the thumb and forefinger at the holes, sear nose up and to the rear; then, taking the long end of the sear release stop lever in the right hand, short end up and to the front, inserting its short end upward through the slot in the forward part of the sear and alining the holes. Replace sear and sear release stop lever on sear carrier with sear nose up and to the rear. Replace sear pin (27) by alining the holes in the sear release stop lever, sear, sear carrier, and trigger guard with the recoil spring guide and inserting the sear pin as the recoil spring guide is slowly worked out (considerable pressure must be exerted by means of the recoil spring guide in order to overcome the resistance of the strong counterrecoil spring). The sear pin must be replaced before the trigger pin in order that tension of the counterrecoil spring will always be on the sear pin.

(2) Replace connector (26) in the trigger (25) by holding the trigger in the left hand with the thumb and forefinger at the hole and the toe of the trigger in the palm of the hand. Pick up the connector in the right hand, toe to the rear and down, and place it in its seat in the trigger. Place the trigger and connector in the trigger guard by holding the trigger guard in the palm of the right hand with the sear carrier up, forward end of trigger guard tipped downward at about a 45° angle, and placing the trigger and connector down through the sear carrier until the connector passes un-

der and to the rear of the connector stop. Replace trigger pin (24). Place the two outer leaves of the sear spring on top of the forward shoulders of the sear and the middle leaf under the forward end of the sear release stop lever, depress the rear end of the sear and slide the sear spring to the rear until its shoulders are in the rear of the slots in the trigger guard, press down and forward on the sear spring with the thumb of the right hand until the front end of the sear spring (23) rests in the depression stop.

(3) Replace magazine release (22). Replace magazine catch (21), spring depression facing out. Replace magazine catch pin (20). Replace magazine catch spring (19). Insert ejector (18) into recess and move it down until it is flush with the magazine catch spring. Compress magazine catch spring in its seat and move ejector down until it is flush with the ejector lock. Depress the ejector lock with the base of a dummy cartridge and push ejector home until the ejector lock is in its proper position.

(4) After trigger mechanism has been assembled, turn change lever to forward position, and pull trigger. If connector will not rise, it is not in place correctly. It should rise and snap out from under sear. If connector will rise but does not raise sear, the middle leaf of the sear spring is weak and the sear spring should be replaced.

c. Operating group (figs. 1 and 2).—Replace extractor spring (16) in the extractor with the short end in the hole and the long end along the groove in the side. Place extractor (15) in its seat in the bolt. Replace bolt link (13) in the bolt lock with the shoulder of the link against the flat surface of the bolt lock. Replace bolt link pin (12). Lift bolt lock and insert the firing pin (11) in its receptacle in the bolt. Lay rifle barrel down and pointing to the left so that rifle is resting on the front sight cover and rear sight base. With bolt (14) and its assembly held in a perpendicular position, insert it in the receiver between the bolt guide and the rear ends of the bolt supports. Press the assembled bolt down so it will lie flat in its place. With bolt link down, push bolt forward. Replace slide (10) with the sear notch up, and push it back all the way. With hammer (9) held with the thumb and forefinger of the left

hand, place it in its seat in the slide, rounded end to the rear. Push bolt lock fully into its locking recess and push slide forward. Using the recoil spring guide as a drift, align the hammer pin holes of the bolt link, hammer, and slide with the hammer pin holes in the sides of the receiver. Insert hammer pin (8) to the right, allowing about $\frac{1}{4}$ inch of the hammer pin to protrude from the receiver. Replace operating handle (7). Push protruding hammer pin all the way home. Replace recoil spring (6). Insert recoil spring guide (5) in the recoil spring, and, with the end of the index finger on the checkered end of the recoil spring guide, turn it until it is properly seated. Replace trigger group (4). Replace trigger guard retaining pin and spring (3). Cock the rifle. Slide gas cylinder tube and forearm (2) to the rear over the gas piston until its rear end is seated in the slot in the front end of the receiver. Replace the gas cylinder tube retaining pin and spring (1). Test rifle by pulling the trigger.

■ 10. TO REMOVE FIRING PIN WITHOUT DISASSEMBLING (fig. 5).—To remove firing pin without disassembling the rifle, lay rifle upside down on table, muzzle to the front. Remove the trigger group. Place rim of cartridge under *center* of bolt guide. With the right hand, slowly pull the operating handle all the way to the rear, taking care not to dislodge the rim of the cartridge from beneath the bolt guide. With the left hand, palm up, grasp the stock so that at least two fingers of the left hand are on the operating handle, holding it to the rear, while the thumb of the left hand is on the bolt link in the rear of the receiver (fig. 5 ①). Release the operating handle with the right hand, it still being held in the rearmost position by the fingers of the left hand. Pry the bolt guide outward with the right hand by placing the middle finger of the right hand under the base of the cartridge while the thumb exerts pressure on the nose (fig. 5 ②). Press down on the bolt link with the thumb of the left hand, causing the bolt to break at the bolt lock pin (fig. 5 ③). Allow bolt to go forward until it stops. Change firing pin. Pull operating handle to the rear again and push bolt into position.

■ 11. TO REMOVE AND REPLACE EXTRACTOR WITHOUT DISASSEMBLING.—*a. Removal* (fig. 6).—Pull operating handle toward the rear and insert an *empty* cartridge case, base to

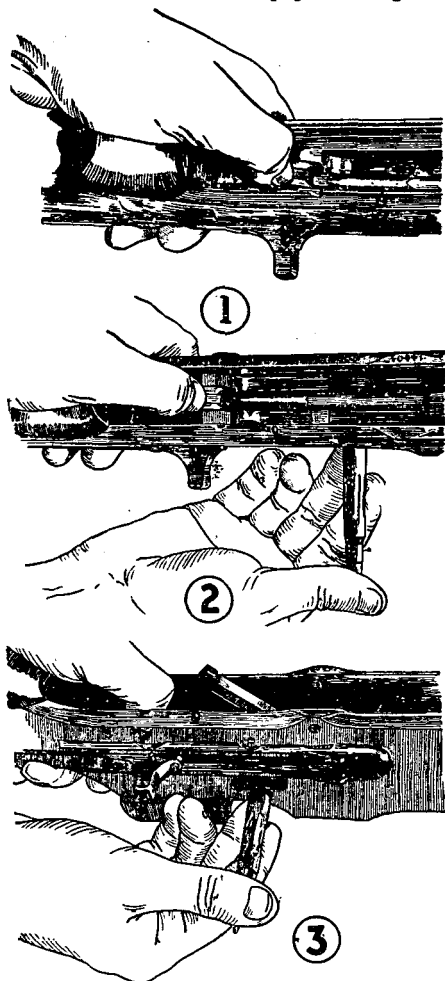


FIGURE 5.—Removing firing pin without disassembling.

the front, between the bolt and chamber, exposing the extractor in the ejection opening (fig. 6 ①). Lay rifle on its side so that the ejection opening is up. With forefinger of

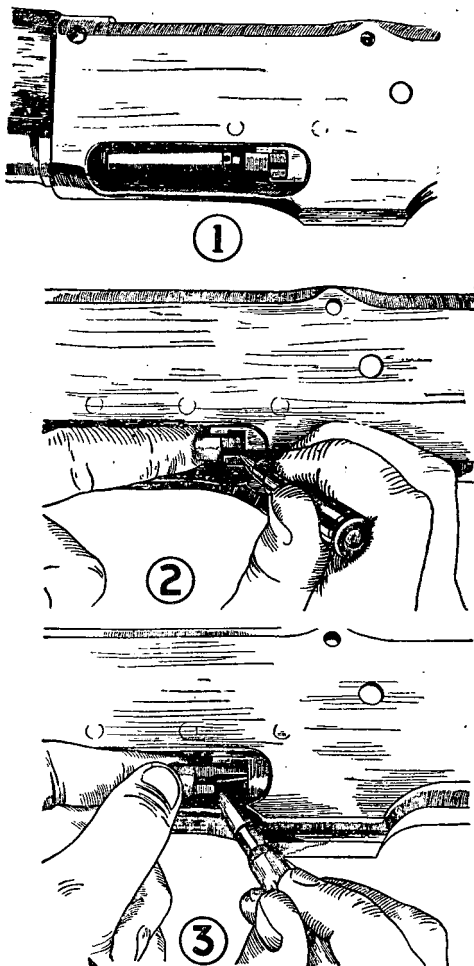


FIGURE 6.—Removing extractor without disassembling.

the left hand, pull out claw of the extractor, then place nose of a cartridge behind the extractor shoulder (fig. 6 ②) and pry it forward until extractor is free of the recess (fig. 6 ③). Remove extractor and extractor spring.

b. Replacement.—Insert short end of extractor spring in the hole in the shank of the extractor so that the long end on the spring is along the slot in the extractor. Insert extractor and spring in the end of the bolt and push them into position. Remove empty cartridge case.

■ 12. **DISASSEMBLING AND ASSEMBLING MAGAZINE** (fig. 7).—Raise rear end of magazine base until indentations are clear, then slide it to the rear. The magazine spring and follower will then fall out. It is assembled in the reverse order.

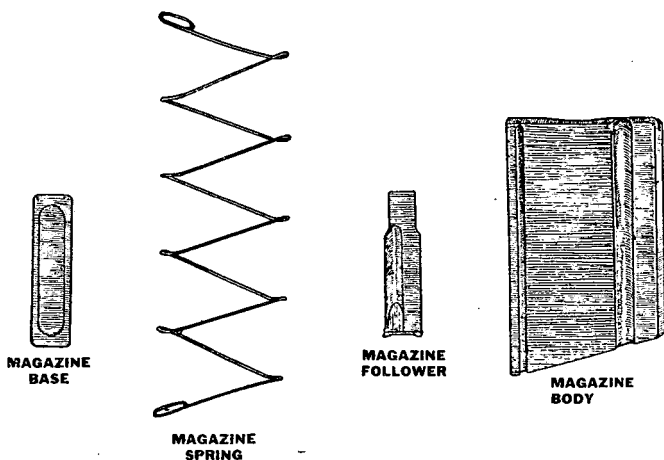


FIGURE 7.—Magazine.

SECTION III

CARE AND CLEANING

■ 13. **GENERAL.**—*a. Importance.*—Care and cleaning includes the care of the automatic rifle necessary to preserve its condition and appearance under all conditions at all times. Unserviceable rifles are usually the result of improper care

and cleaning. Rifles in the hands of troops should be inspected daily to insure proper condition and cleanliness.

b. Lubricants, cleaning materials, and rust preventives.—The following are the only materials authorized and issued for the care and cleaning of the Browning automatic rifle. The use of unauthorized materials such as abrasives is prohibited.

(1) *Cleaner, rifle bore.*—(a) Rifle bore cleaner is issued for cleaning the bore of the rifle after firing. It possesses rust-preventive properties and provides temporary protection against rust. It is preferable, however, to dry the bore immediately after cleaning and oil it lightly with light preservative lubricating oil.

(b) Rifle bore cleaner will freeze at temperatures below 32° F. Therefore, closed containers should not be filled more than three-fourths full in freezing weather as full containers, when frozen, will burst. Once frozen and thawed out, it must be shaken well before using.

(2) *Oil, light preservative, lubricating.*—This oil is used for lubrication of all moving parts and is also effective as a rust-preventive for short term storage of the automatic rifle. Depending on climatic conditions, it is effective for periods of 2 to 6 weeks. However, rifles in short-term storage must be inspected every 4 or 5 days to renew the preservative film, if necessary. For longer periods of storage, the rifles will be protected with light rust-preventive compound.

(3) *Oil, lubricating, for aircraft instruments and machine guns.*—This oil may be used for lubricating the rifles when light preservative oil is not available or in temperatures below 32° F. It is an extremely light oil which relies entirely upon maintenance of a film to protect metal surfaces from rusting. When used as a preservative, the metal parts must be inspected daily for rust, cleaned, and again lightly coated with the oil.

(4) *Oil, engine, SAE 10 or other light machine oil.*—Any good, light oil may be used when oils previously mentioned can not be obtained. However, in cold weather any of the heavier oils will cause sluggish operation or rifles may not work at all. Alternate oils do not possess the rust-preventive

properties of preservative lubricating oil, and when used, matériel should be examined, cleaned, and recoiled frequently.

(5) *Compound, light rust-preventive*.—This compound is issued for protecting metal parts for long periods of time while rifles are boxed and in storage. It should be warmed before application. However, when heating facilities are not available, it can be effectively brushed onto the parts, provided the temperature is above 80° F. Below 80° F., light rust-preventive compound becomes sluggish and its use without preheating is uneconomical.

(6) *Solvent, dry-cleaning*.—This is a petroleum distillate and therefore inflammable. Smoking is prohibited where it is employed. Dry-cleaning solvent is used for removing grease, oil, or rust-preventive compound. Metal surfaces are bathed in the solvent with rag swabs. After bathing, the solvent must be thoroughly wiped off all metal surfaces with clean rags. After cleaning with solvent, metal parts are especially susceptible to corrosion from finger marks. Finger marks must be wiped off. Cleaning solvent will attack and discolor rubber.

(7) *Soap and water*.—If rifle bore cleaner is not available, water will be used; warm water is good, but warm, soapy water is better. It is used to clean the bore and gas system.

(8) *Soda ash*.—Soda ash, a commercial sodium carbonate, may be used in water in lieu of soap.

(9) *Oil, linseed*.—Raw linseed oil is applied to the woodwork of the rifle at intervals to preserve the wood and to keep out moisture.

(10) *Oil, neat's-foot*.—This oil is used for cleaning and softening the sling. It should be used sparingly, rubbed in, and wiped off carefully to prevent soiling clothing.

■ 14. CARE AND CLEANING.—*a. General*.—(1) Automatic rifles and magazines in the hands of troops and in temporary storage within the unit must receive daily attention.

(2) Automatic rifles will be disassembled only to the extent necessary for proper cleaning. Whenever necessary, magazines will be disassembled for cleaning and light lubrication.

(3) Proper oiling is second in importance only to intelligent cleaning. It is a vital necessity for the working parts,

but the oil should be used sparingly; wiping with a well-oiled rag is the best method.

b. When no firing is being done.—Care and cleaning of the automatic rifle and magazines when no firing is being done will include the care necessary to preserve their condition and appearance.

(1) To clean the bore of the automatic rifle, pass a dry patch through the bore by inserting the cleaning rod through the muzzle end. Repeat with dry patches, unless the bore is unduly dry in which case an oily patch will be used to loosen the dry film, until several successive patches have come out clean. Push a patch dampened with oil through the bore to protect its surface. Dust out all screw heads and crevices with a small cleaning brush or wood splinter. Wipe all metal surfaces with a dry cloth to remove moisture, perspiration, and dirt. Wipe the outer surfaces of the automatic rifle, including the forearm, with a lightly oiled cloth (see par. 13b for proper lubricants). This protective film on all metal parts will be maintained at all times. At least once a month, and always after the stock and forearm have become wet, they should be rubbed thoroughly with a little linseed oil in the palm of the hand. Rub oil in until dry. Use neat's-foot oil moderately for cleaning and softening the sling.

(2) It is imperative that magazines be given the best of care and kept in perfect condition. They must be kept free from dirt and rust and handled so as to avoid any dents or abrasions. To clean, they should be disassembled when necessary, wiped clean and dry, and thinly coated with oil.

c. After firing.—(1) The bore of the rifle will be thoroughly cleaned by the evening of the day on which it is fired and similarly cleaned for the next three days.

(2) Disassemble the rifle. The bore is cleaned by swabbing with a patch saturated with rifle bore cleaner. Repeat with several patches. If rifle bore cleaner is not available, water will be used. The water should be soapy and preferably warm. Soda ash may be used in lieu of soap. While swabbing the bore with water, rest the rifle on the toe of the butt, muzzle elevated slightly so that the water drips out of the receiver and does not get into the buffer, rate-reducing mechan-

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ism, and the locking recess. While the bore is still wet, run the metal brush through it several times. Remove the brush and swab the bore again with a patch saturated with cleaner or with water. Follow this with dry patches until several patches come out clean and dry. When it is clean, push a patch dampened with oil through the bore to protect its surface.

(3) Clean the chamber with the chamber cleaning brush, wipe clean with a cloth, and oil lightly.

(4) To clean the gas system, remove the gas cylinder from the gas cylinder tube. Insert the smooth end of the body of the gas cylinder cleaning tool into the gas cylinder. Turn it to the right while gradually scraping into the cylinder head. As it reaches the head, apply additional pressure to the tool and give it a few turns to cut the carbon from the bottom inside surface. Withdraw and reverse the tool. Using the recess cutter as a gage, remove the carbon from the recesses at the forward end of the interior of the gas cylinder. With the drift point, clean the gas ports in the barrel, gas cylinder tube, and gas cylinder. Scrape the carbon from the face of the piston with the front cutting edge of the tool body, and remove the deposit from between the piston rings with the drift point. Wipe thoroughly with a rag saturated with rifle bore cleaner, if available. Otherwise, wash with hot water and soap or soda ash solution (if not available, use plain water) and dry thoroughly. Inspect for cleanliness. When it is clean, oil lightly. Clean the gas cylinder tube by washing with hot water and soap or soda ash solution (if unavailable, use plain hot or cold water). This cleaning may be facilitated by tying a cloth tightly into the center part of a string or thong and pulling it back and forth inside the tube. Dry thoroughly and *inspect*. When clean, oil lightly.

(5) Wipe the remaining parts of the operating group dry and clean; apply a thin film of oil.

(6) Wipe the trigger group dry, disassembling it only periodically or when necessary, and oil lightly.

(7) Wipe the outer metal parts of the rifle dry and apply a thin film of oil.

(8) Clean the stock and forearm of the rifle and wipe the sling with a clean, dry cloth.

(9) Inspect the interior of the magazines by depressing the follower with the thumb. If the interior is dirty, disassemble, clean thoroughly, and oil lightly. Otherwise, merely wipe clean and dry, and oil. Much magazine trouble can be avoided by proper handling to avoid dirt and dents. It is imperative that magazines be given proper care and handling.

d. Points to be observed before, during, and after firing.—See paragraph 40.

■ 15. STORAGE.—For preserving and lubricating materials, see paragraph 13b.

a. Preparation for storage.—Automatic rifles should be cleaned and prepared for storage with particular care. The bore, all parts of the mechanism, and the exterior of the rifles should be thoroughly cleaned and then completely dried with rags. In damp climates, particular care must be taken to see that the rags are dry. After drying a metal part, the bare hands should not touch that part. Special care should be taken to insure that the gas system is thoroughly cleaned and that the gas ports are free from fouling. All metal parts should then be coated with either light preservative lubricating oil, or light rust-preventive compound, depending on the length of storage required. Before placing the automatic rifle in the packing chest, the wooden supports for the butt and muzzle will be painted with rust-preventive compound. Under no circumstances will a rifle be placed in storage while in a cloth or other cover, or with a plug in the bore. Such articles collect moisture which causes the weapon to rust.

b. Cleaning when received from long-term storage.—Automatic rifles which have been stored in accordance with *a* above will be coated with either light preservative lubricating oil or with rust-preventive compound. Weapons received from ordnance storage will in general be coated with rust-preventive compound. Use dry-cleaning solvent to remove all traces of the compound or oil, particular care being taken that the gas system, gas ports, firing pin, and all recesses in which springs or plungers operate are cleaned thoroughly.

After using the cleaning solvent, make sure it is completely removed from all parts by wiping with a dry cloth.

NOTE.—Failure to clean the gas system, the firing pin, and the recesses of the bolt in which it operates may result in gun failure at normal temperatures and will certainly result in serious malfunctions if the rifle is operated in low temperature areas, as rust-preventive compound and other foreign matter will cause the lubricating oil to congeal on the mechanism.

■ 16. CARE AND CLEANING UNDER UNUSUAL CONDITIONS.—*a. In cold climates.*—(1) For temperatures below freezing, it is necessary that the moving parts of the weapon be kept absolutely free from moisture. It has also been found that excess oil on the working parts will solidify to such an extent as to cause sluggish operation or complete failure. Oil must be applied *very* sparingly by wiping with a cloth which has first been wet with light preservative lubricating oil and then wrung out.

(2) The metal parts of the weapon should be taken apart and completely cleaned with dry-cleaning solvent or equivalent before use in temperatures below 0° F. The working surfaces of parts which show signs of wear may be lubricated by rubbing with a cloth slightly dampened with light preservative lubricating oil as prescribed in (1) above. At temperatures above 0° F., all metal surfaces of the weapon may be oiled lightly after cleaning, by wiping with a slightly oiled cloth, using light preservative lubricating oil.

(3) When it is brought indoors, the weapon should first be allowed to come to room temperature. It should then be disassembled, wiped completely dry of the moisture which will have condensed on the cold metal surfaces, and thoroughly oiled with light preservative lubricating oil. This will be done every time a weapon is carried into a warm room after exposure to low temperature. If possible, such condensation should be avoided by providing a cold place in which to keep rifles when not in use. For example, a separate cold room with appropriate racks may be used, or in the field, racks under proper cover may be set up outdoors.

(4) If the weapon has been fired, it should be thoroughly cleaned and oiled. The bore may be swabbed out with an oily patch and when the weapon reaches room temperature, thoroughly cleaned and oiled as prescribed in paragraph 14c.

(5) Before firing, the weapon should be cleaned and oil removed as prescribed in (2) above. The bore and chamber should be entirely free of oil before firing. In cold climates, the medium gas port will usually be found to be the most satisfactory initial setting of the gas cylinder.

b. In hot climates.—(1) *Tropical climates.*—(a) Light preservative lubricating oil should be used for lubrication.

(b) In tropical climates where temperatures and humidity are high, or where salt air is present, and during rainy seasons, the weapon should be thoroughly inspected daily and kept lightly oiled when not in use. The groups should be dismounted at regular intervals and, if necessary, disassembled sufficiently to permit the drying and oiling of parts.

(c) Care should be exercised to see that unexposed parts and surfaces are kept clean and oiled.

(d) Wood parts should also be inspected to see that swelling due to moisture does not bind working parts. (In such cases shave off only enough wood to relieve binding.) A light coat of raw linseed oil applied at intervals and well rubbed in with the heel of the hand will help to keep moisture out. Allow oil to soak in for a few hours and then wipe and polish wood with a dry clean rag.

NOTE.—Care should be taken that linseed oil does not get into mechanism or on metal parts as it will gum up when dry.

(2) *Hot, dry climates.*—(a) In hot, dry climates where sand and dust are likely to get into the mechanism and bore, the weapon should be wiped clean daily or oftener if necessary. Groups should be dismounted and disassembled as far as necessary to facilitate thorough cleaning.

(b) When the weapon is being used under sandy conditions, the weapon should be wiped completely dry of all lubricant. This will prevent sand carried by the wind from sticking to the lubricant and forming an abrasive compound which will ruin the mechanism. Immediately upon leaving sandy terrain, the weapon must be relubricated with light preservative lubricating oil.

(c) In such climates, wood parts are apt to dry out and shrink, and a *light* application of linseed oil, applied as in (1) (d) above, will help to keep wood in condition.

(d) Perspiration from the hands is a contributing factor

to rust because it contains acid, and metal parts should be wiped dry frequently.

(e) During sand or dust storms, the breech and muzzle should be kept covered.

c. Care when subject to gas attack.—(1) It is important to prevent chemicals used in a gas attack from coming into contact with the automatic rifle, ammunition, and accessories. When a gas attack is anticipated, the following action will be taken: apply oil to all outer metal surfaces of the automatic rifle and accessories. If it is not to be used, cover rifle, accessories, and ammunition. *Do not apply oil to ammunition.*

(2) After the attack, determine by means of detector crayon or paper whether or not the matériel is contaminated (par. 127, FM 21-40). If uncontaminated, clean matériel with dry-cleaning solvent. Prepare for use as required.

(3) If contaminated, the following action will be taken:

(a) A suit of protective clothing (permeable or impermeable) and a service gas mask must be worn for decontamination.

(b) Matériel contaminated with chemicals other than mustard or lewisite must be cleaned as soon as possible with dry-cleaning solvent or denatured alcohol.

(c) Do not allow the chemical agent to come into contact with the skin. Burn, or *preferably bury*, all rags or wiping materials used for decontamination. *Extreme caution should be taken to protect men against fumes created by burning.*

(d) If the surface of the matériel is coated with grease or oil and has been in a mustard or lewisite attack, first remove the grease or oil by wiping with rags wet with dry-cleaning solvent.

(e) Decontaminate the rifle with a solution of agent, decontaminating, noncorrosive (par. 132, FM 21-40). Prepare this by mixing one part of agent, decontaminating, noncorrosive, to 15 parts of solvent (acetylene tetrachloride) by weight, or by mixing one part of decontaminating agent, noncorrosive, with six parts of solvent by volume.

(f) After decontamination, clean the matériel thoroughly and prepare for use as required.

(g) Detailed information on decontamination is contained in FM 21-40, TM 9-850, and TM 3-220.

SECTION IV

FUNCTIONING

■ 17. GENERAL.—*a. Object.*—This section is designed to provide a nontechnical description of the functioning of the rifle. The object of instruction in this subject is to lead the soldier to an understanding of the simple functioning of his weapon without emphasis on memorizing the matter of the text.

b. When taken up.—Instruction in functioning will be taken up after instruction in the disassembly, assembly, care and cleaning of the rifle.

■ 18. USE OF DUMMY CARTRIDGES.—The corrugated type of dummy cartridge (cal. .30, M1906) may be used for instruction in functioning. The use of the slotted type of dummy cartridge (range, cal. .30, M1) is prohibited. Special care must be exercised in the use of dummy cartridges that they do not introduce dirt or grit into the chamber of the rifle.

■ 19. MECHANICAL MEANS OF FUNCTIONING.—All automatic weapons must have mechanical means for performing the following functions: extraction, ejection, feeding, locking the breech while there is a high pressure in the bore, and igniting the cartridge. Operations such as extraction and ejection are performed by various cams, lugs, and springs, and the energy necessary to perform this work and to overcome friction in the rifle is derived from the explosion of the powder in the chamber.

■ 20. CYCLE.—*a.* One rearward and one forward movement of the mechanism is a cycle. This begins with the ignition of the cartridge in the chamber and ends at the ignition of the succeeding cartridge. The functioning of the automatic rifle is divided into two phases based on the operation of the mechanism in one cycle. These two phases are the rearward movement and the forward movement. The striking of the buffer by the slide marks the division of the two phases.

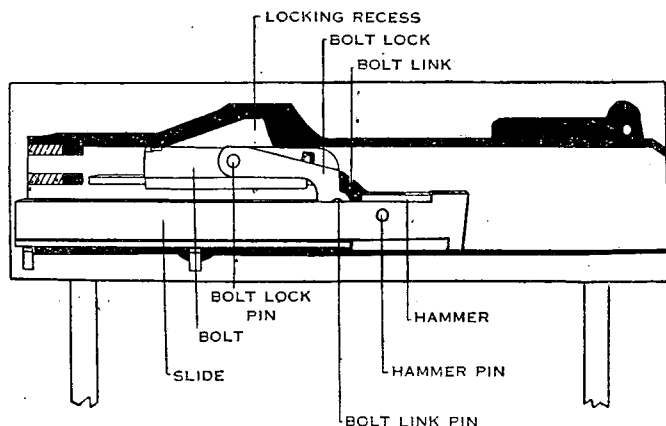


FIGURE 8.—Wooden working model showing bolt action.

b. The operations which take place in the rearward movement are—

- (1) Action of gas.
- (2) Movement of slide to rear.
- (3) Unlocking.
- (4) Withdrawal of firing pin.
- (5) Extraction.
- (6) Ejection.
- (7) Termination of first phase.

c. The operations which take place in the forward movement are—

- (1) Action of recoil spring.
- (2) Feeding.
- (3) Locking.
- (4) Ignition.
- (5) Termination of second phase.

■ 21. DESCRIPTION OF CYCLE.—*a. Rearward movement.*—(1) *Action of gas.*—A cartridge having been ignited, the bullet, under the pressure of the expanding powder gases, travels through the barrel, and when it reaches a point 6 inches from the muzzle it passes a port in the bottom of the barrel. The barrel pressure, which at this instant is still very high, seeks

this first natural vent. Alined with the barrel port are other similar ports in the gas cylinder tube bracket, gas cylinder tube, and gas cylinder. The port in the gas cylinder is the smallest and serves to throttle the barrel pressure. The port in the gas cylinder leads radially into a well about $\frac{1}{8}$ inch in diameter in the head of the gas cylinder. The throttled barrel pressure is conducted through this well to the gas piston plug. This pressure acts on the piston for the very short time which it takes for the bullet to travel the 6 inches of distance from the barrel port to the muzzle. Its effect is that of a sudden severe blow on the piston plug. Under the impact of this blow, the gas piston is driven to the

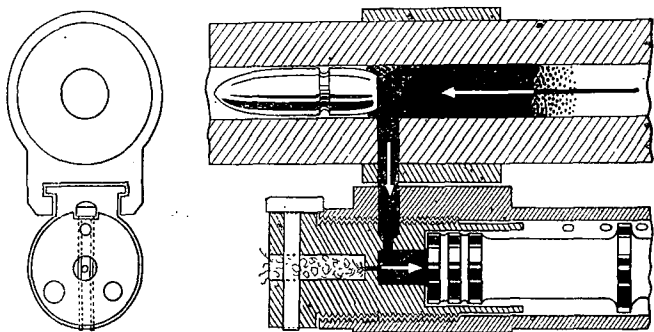


FIGURE 9.—Action of gas.

rear carrying the slide with it. When the piston has traveled about $\frac{9}{16}$ inch backward, the bearing rings on its head and the gas piston plug pass out of the cylinder. The gas expands around the piston head and into the gas cylinder tube and is exhausted through the six portholes in the tube. The gas is prevented from traveling back through the gas cylinder tube by the two rings on the piston, about $\frac{5}{8}$ inch apart and $1\frac{1}{4}$ inches from the piston head. These rings also serve as bearings to hold the front end of the piston in the center of the gas cylinder tube after the piston head has passed out of the gas cylinder.

(2) *The slide.*—As the piston is forced back, it carries the slide with it. The first and the immediate result of the back-

ward movement of the slide is to begin the compression of the recoil spring, thereby storing energy for the forward action. The hammer pin is slightly in advance of the bolt link pin, about 0.19 inch. The center rib of the hammer is very slightly in rear of the head of the firing pin. When the slide begins its motion to the rear, it imparts no motion whatever to the bolt and bolt lock. The slide moves back 0.19 inch, and its only effect during this travel is to carry the hammer from the firing pin and the hammer pin directly under the bolt link pin.

(3) *Unlocking.*—At this point the unlocking begins. The bolt link revolves forward about the hammer pin, drawing the bolt lock down and to the rear. The motion of the lock and bolt, which is zero at the instant the hammer pin passes under the bolt link pin, accelerates from this point until the slide has traveled about 1.2 inches, at which latter point the bolt lock is drawn completely down out of the locking recess and away from the locking shoulder of the receiver. It is now supported in front on the bolt supports. The front upper surface of the bolt link has revolved forward and bears upon the locking surface of the bolt lock. These two influences prevent the bolt lock from revolving down below the line of backward travel of the bolt.

(4) *Withdrawal of firing pin.*—As the bolt lock revolves down from its locked position, a cam surface in a slot in the rear bottom side of the bolt lock comes in contact with a similar cam surface on the firing pin lug. This action cams the firing pin from the face of the bolt.

(5) *Extraction.*—The backward motion of the bolt begins when the bolt lock has been drawn down so that the circular cam surface on its under side is operating on the rear shoulders of the bolt supports. This produces a strong lever action which slowly loosens the cartridge case. The backward travel of the bolt has been slight, only about $\frac{5}{32}$ inch when the firing pin is withdrawn; its travel is about $\frac{11}{32}$ inch when the bolt lock is completely drawn down. From this point the bolt moves to the rear, drawn by the bolt lock and bolt link, with the same speed as the slide and carries with it the empty cartridge case which is held firmly in its seat on the face of the bolt by the extractor. The extractor is on the

upper right side of the bolt next to the ejection opening in the receiver. A slot cut in the left side of the bolt lock near the back end passes over the bolt guide, which supports the bolt lock and bolt when they are in the rear position.

(6) *Ejection*.—When the slide reaches a point about $\frac{1}{4}$ inch from the end of its travel, the base of the cartridge case strikes the ejector. This action causes the cartridge case to be pivoted with considerable force about the extractor and through the ejection opening in the receiver. The front end of the cartridge case passes first out of the receiver and is pivoted so that it strikes the outside of the receiver at a point about 1 inch in rear of the ejection opening. It rebounds from the receiver toward the right front.

(7) *Termination of rearward movement*.—The rearward motion terminates when the rear end of the slide strikes the buffer head and sear release. The slide, under the action of the recoil spring, moves forward $\frac{1}{10}$ inch after striking the buffer head and sear release. If the sear nose is not depressed, it engages the sear notch on the slide and the piece is cocked for the next burst or shot.

NOTE.—The motion of the bolt, bolt lock, and bolt link mechanism begins slowly at first and does not attain the speed of the slide until the latter has traveled about $1\frac{1}{4}$ inches backward. This is a very important characteristic of the rifle, since on this account the mechanism is not subjected to an excess strain due to a sudden start at the instant the gas strikes the piston. This slow start also allows the bullet to clear the muzzle thereby preventing a blow-back of the expanding gases.

b. *Forward movement*.—(1) *Action of recoil spring*.—When the trigger is pressed, the sear nose is depressed, disengaging the sear, and the slide moves forward under the action of the recoil spring. The position of the bolt link pin is slightly below a line joining the bolt lock pin and the hammer pin; therefore, as the slide starts forward, the joint at the bolt link pin has a tendency to buckle downward. It is prevented from doing this by the tail of the feed rib on the bolt which extends backward under the bolt lock, also by the upper front surface of the bolt link being in contact with the locking surface of the bolt lock. Since the joint cannot buckle, the entire mechanism moves forward with the slide. When it has traveled about $\frac{1}{4}$ inch, the front end of the feed rib strikes

on the base of the top cartridge, which the magazine spring and lips are holding up in its path.

(2) *Feeding* (fig. 10).—The cartridge is carried forward about $\frac{1}{4}$ inch, when the nose of the bullet strikes the bullet ramp or guide on the breech of the barrel and is deflected upward toward the chamber. This action also guides the front end of the cartridge from under the magazine lips. The base of the cartridge approaches the center of the magazine, where the lips are cut away and the opening enlarged, and at this point is forced out of the magazine by the magazine spring. The base of the cartridge slides up the face of the bolt and under the extractor. Should the cartridge fail to

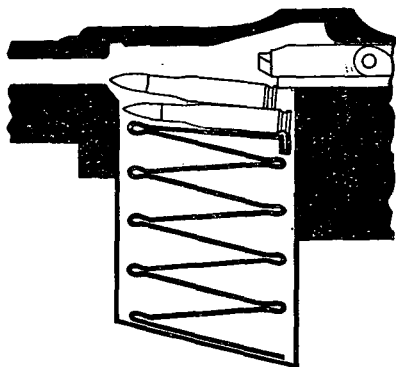


FIGURE 10.—Feeding.

slide under the extractor, the extractor will snap over its head as the bolt reaches the forward position. When the cartridge is released by the magazine, the nose of the bullet is so far in the chamber that it is guided by the chamber from this point on.

(3) *Locking*.—When the slide is about 2 inches from its forward position, the circular cam surface on the under side of the bolt lock begins to ride over the rear shoulders of the bolt supports on the sides of the receiver and the rear end of the bolt lock is cammed upward. The bolt link pin passes up above the line joining the bolt lock pin and hammer pin. The joint at the bolt link pin now has a tendency

to buckle upward, and the bolt lock being opposite the locking recess in the receiver is free to and does pivot upward about the bolt lock pin. The bolt link revolves upward about the hammer pin, forcing the bolt lock up, and a rounded surface on the bolt lock just above the locking face slips over the locking shoulder in the receiver, giving the lock a lever action which forces the bolt home to its final position. The two locking surfaces on the bolt lock and the receiver register as the hammer pin passes under the bolt link pin.

(4) *Igniting cartridge.*—The lug on the firing pin is buried in the slot on the under side of the bolt lock at all times except when the bolt is locked in the forward position. Therefore, the firing pin is locked away from the face of the bolt during all the rearward and forward motion of the bolt. When the hammer pin passes under the bolt link pin, which places the bolt lock into the locking recess in the top of the receiver, the firing pin has been released by the bolt lock. The slide and hammer move forward about $\frac{1}{10}$ inch farther, and the center rib of the hammer strikes the head of the firing pin a smart blow, thereby igniting the cartridge.

(5) *Termination of second phase.*—The forward end of the slide strikes a shoulder at the rear end of the gas cylinder tube, terminating the forward motion. The forward motion is not terminated by the hammer on the firing pin. This can be seen by examining the head of the firing pin when the gas cylinder tube is assembled to the receiver, and the bolt mechanism is in the forward position. The firing pin has still about $\frac{1}{16}$ -inch clearance from its extreme forward position.

NOTE.—The locking shoulder of the receiver is inclined forward. Its surface is perpendicular to the line through the bolt lock which the shock of the explosion follows; therefore the force of this shock is exerted squarely against the normal surface. The speed of the bolt mechanism is slowed down gradually from the instant that the bolt lock starts to rise until the hammer pin passes under the bolt link pin, when the speed is zero.

■ **22. FUNCTIONING OF COMBINATION BUFFER AND RATE REDUCING MECHANISM (fig. 11).**—*a. Parts.*—The buffer and rate reducing mechanism consists of the following parts in the order given from front to rear:

(1) Buffer tube, which has four splined slots on the inside of the forward end for use in positioning the buffer head.

One of these slots only is used to anchor the buffer tube to the buffer head by means of the buffer key.

- (2) Buffer head.
- (3) Sear release, inserted in buffer head.
- (4) Buffer key, which holds sear release in buffer head and anchors buffer tube to buffer head.
- (5) Brass friction cup with concave interior which is split to allow for expansion; and a steel cone to fit into the cup.

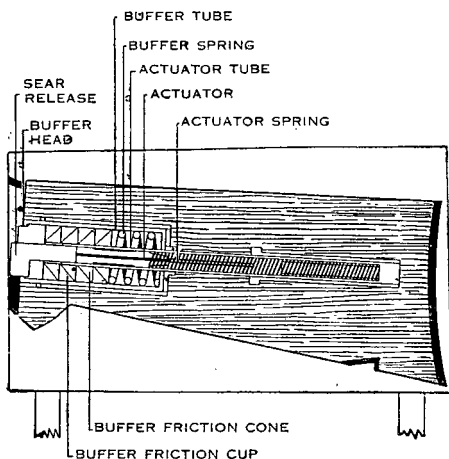


FIGURE 11.—Wooden working model of the combination buffer and rate-reducing mechanism.

Four of these cups and cones are placed one after the other in series.

- (6) Buffer spring.
- (7) Actuator, inside of cups, cones, and buffer spring.
- (8) Buffer tube cap.
- (9) Actuator tube.
- (10) Actuator spring, inside actuator tube.
- (11) Stock retaining sleeve lock washer.
- (12) Stock retaining sleeve.
- (13) Actuator stop.
- (14) Pin.

b. Action.—(1) When the automatic rifle is fired at the slow cyclic rate, the buffer head and sear release, struck by the rear end of the slide, move to the rear. The buffer head forces the friction cups over the cones, causing them to expand tightly against the tube, consequently producing considerable friction as the cups move back and compress the buffer spring. Thus the rearward action of the slide is checked gradually and there is practically no rebound. The buffer spring returns the buffer head and friction cups and cones to their original positions. After striking the buffer head and sear release, the slide moves forward until it is engaged by the sear. The delayed release of the sear is then accomplished as follows: The sear release when struck by the slide in turn strikes the actuator in the buffer. The actuator is driven to the rear inside the actuator tube against the actuator spring. It returns under the force of expansion of the actuator spring to drive the sear release forward against the camming surface on the rear of the sear, thus releasing the sear and permitting the rifle to continue its cycle.

(2) When the rifle is fired at the normal cyclic rate, the functioning of the combination buffer and rate reducing mechanism is the same as explained for the slow cyclic rate in (1) above except that the sear release is prevented from engaging the camming surface on the rear end of the sear by the sear release stop lever.

■ **23. FUNCTIONING OF TRIGGER MECHANISM.**—*a.* The trigger mechanism (fig. 12) has three settings:

(1) *Normal cyclic rate (A).*—When so set, the sear is depressed as long as the trigger is held back, and the rifle will continue firing at a cyclic rate of about 550 rounds per minute until the magazine is emptied.

(2) *Slow cyclic rate (F).*—When so set, the sear is depressed, thereby disengaging the sear from the sear notch on the under side of the slide when the trigger is pulled. However, the mechanism is so constructed that the sear rises and engages in the sear notch when the slide comes back again, and the sear notch will not disengage until the sear release in the buffer depresses the rear end of the sear by riding over

the cammed end on the sear or until the trigger is fully released and then pressed. With this setting, the rifle will fire at the slow cyclic rate of 350 rounds per minute for the length of time the trigger is held completely to the rear, or one shot may be fired for each press and quick release of the trigger.

(3) *Safe (S).*—When so set, the sear cannot be released from the sear notch by pressing on the trigger.

b. The action of the trigger mechanism is taken up in phases and should be followed on the mechanism itself as the explanation proceeds. Have the trigger group disas-

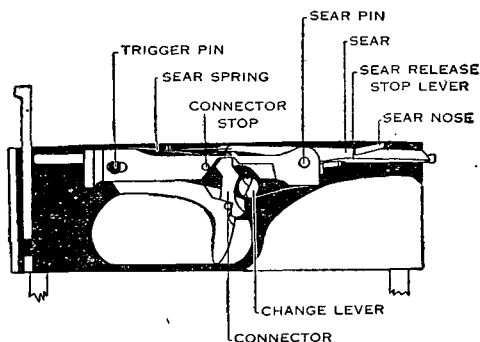


FIGURE 12.—Wooden working model showing trigger action.

sembled completely. Study the shape of the change lever and note the following:

- (1) It is a bar about $\frac{1}{4}$ inch in diameter.
- (2) It has three shallow longitudinal slots cut on top of the bar as the handle is held vertically.
- (3) The side of the bar is slotted in such a way as to leave a little tongue of metal in the center of the lower edge of the slot.

■ 24. SETTING CHANGE LEVER.—*a.* Assemble the change lever and spring to the trigger guard. The toe of the change lever spring is seated in one of the longitudinal slots on the change lever, and as the lever is turned from one position to another it seats in the other slots. The only function

of the forward end of the spring and the longitudinal slots is to hold the change lever in the position in which it is set.

b. Assemble the trigger and pin to the guard.

c. Turn the change lever to rear or safe position. In this position the slot is turned slightly upward, and the full surface of the bar is on the bottom. Pull the trigger. The rear top end of the trigger is slotted longitudinally, and the metal on each side of the slot forms two shoulders which rise against the bottom of the change lever bar.

d. Push the change lever over to the vertical position, which is the normal cyclic rate setting. Pull the trigger. The slot in the change lever is now turned to the front, and the two shoulders of the trigger, which before engaged the full surface of the change lever bar, now are free to pass up into the slot of the change lever; also the tongue of metal on the bottom of the change lever slot passes through the longitudinal slot in the end of the trigger.

e. Push the change lever forward to the slow cyclic rate position. The slot is now turned partially down and when the trigger is pulled the rear end of the trigger passes up into the change lever slot; also the tongue of metal in the bottom of the change lever slot is now turned back and does not pass through the slot in the end of the trigger as it did in the normal cyclic rate position.

■ 25. FUNCTIONING OF CONNECTOR.—a. Place the connector in its seat in the trigger. Set the change lever at "Safe." The trigger cannot be pulled as the shoulders on the heel of the trigger are against the change lever bar. Since the trigger cannot be pulled, the connector does not rise up through the slot in the sear carrier.

b. Set the change lever at the normal cyclic rate (vertical). Pull the trigger. The connector is now raised and held in a vertical position. The tongue on the change lever engages the toe of the connector as the trigger is pulled and holds the connector firmly in an upright position.

c. Set the change lever at the slow cyclic rate (forward). Pull the trigger. The connector is now raised and tips forward since the tongue on the change lever does not engage the toe of the connector. This tipping forward of the connector is aided by the cammed surface underneath the sear

carrier, and the connector is prevented from tipping too far forward by the connector stop which is the cross pin in the sear carrier.

■ 26. FUNCTIONING OF SEAR.—*a.* Completely assemble the trigger group. Set the change lever at "Safe." With the change lever in this position, as previously stated, the connector does not rise as pressure is applied against the trigger. Since the connector cannot be raised to lift the forward end of the sear which is pivoted in the middle, the sear nose on the rear end of the sear will remain engaged in the sear notch on the slide and the rifle will not fire.

b. Set the change lever at the normal cyclic rate (vertical). Press the trigger. The connector, being raised and held in a vertical position, raises the forward end of the sear, thus depressing the rear end. The rear end of the sear, being depressed, disengages the sear nose from the sear notch in the slide, allowing the operating parts to go forward. The slide will be free to move backward and forward as long as the sear nose on the sear is depressed by holding back on the trigger. The sear release stop lever operates as part of the sear and prohibits the camming action of the sear release on the rear end of the sear.

c. Set the change lever at the slow cyclic rate (forward). Press the trigger. As the trigger is pressed the connector rises, lifts the forward end of the sear, and depresses the rear end, thus disengaging the sear nose from the sear notch and allowing the slide to go forward. The connector is then cammed forward from under the sear, allowing the sear to return to its original position. The sear nose will then engage the slide and hold it momentarily to the rear until it is released by the sear release in the buffer mechanism acting on the cammed surfaces on the rear of the sear (par. 22*b*). This action of the sear release on the cammed surface of the sear will continue as long as the trigger is held, and the rifle will continue to fire at the slow cyclic rate of about 350 rounds per minute.

■ 27. FUNCTIONING OF SEAR RELEASE STOP LEVER.—*a.* With the change lever set at "Safe," the sear release stop lever functions as a part of the sear as outlined in paragraph 26*a*.

b. With the change lever set at the normal cyclic rate, the sear release stop lever functions as a part of and with the sear. As long as the trigger is pressed to the rear, the rear ends of both the sear and the sear release stop lever remain depressed and the rifle fires at the normal cyclic rate.

c. With the change lever set at the slow cyclic rate, as long as the trigger is pressed fully to the rear, the sear nose will rise to engage the slide after each shot; but, since the sear release stop lever remains depressed (held up at the front end by the connector), the sear release cams down the cammed end of the sear, thereby releasing the slide for each

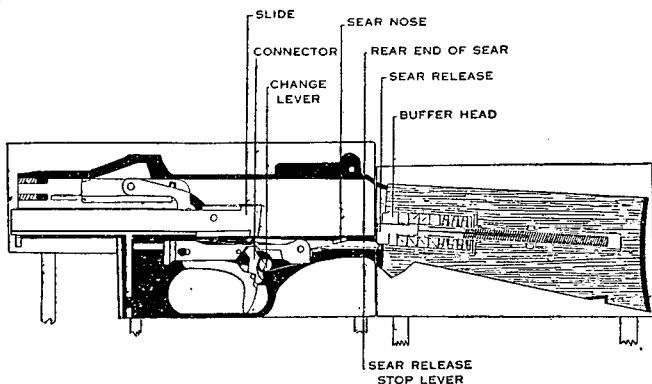


FIGURE 13.—Combined wooden working model to show functioning.

succeeding shot. If the trigger is quickly released as soon as pressed, both sear and sear release stop lever rise, thereby permitting the firing of single shots.

■ 28. FUNCTIONING OF SPRINGS.—The springs in the trigger group operate either to return the parts on which they work to their original positions or to hold them in such positions.

SECTION V

OPERATION

■ 29. GENERAL.—a. *Object.*—This section is designed to give the soldier instruction necessary for the operation of the rifle.

b. When taken up.—The operation of the automatic rifle will be taken up at any convenient time after instruction in care and cleaning (pars. 13 to 16) has been completed.

■ 30. USE OF DUMMY CARTRIDGES.—As prescribed in paragraph 18.

■ 31. TO LOAD THE MAGAZINE.—To load the magazine, place the wide end of the magazine filler over the top of the magazine, so that the groove in the magazine filler fits over the catch rib of the magazine. Hold the magazine in the same relative position that it occupies in the rifle, that is, with the catch rib toward the operator. Then, insert a clip of cartridges in the guides provided in the filler; and, with the right thumb near the base, push the cartridges into the magazine. Each magazine will hold 20 rounds (fig. 14). In case a magazine filler is not available, the magazine may be loaded by hand. This is accomplished by holding the magazine in the left hand, top up, catch rib toward the operator. Insert the cartridges singly in the top of the magazine, base of the cartridge toward the operator, with the thumb of the right hand, pressing downward until the cartridge is held firmly by the lips at the top of the magazine.

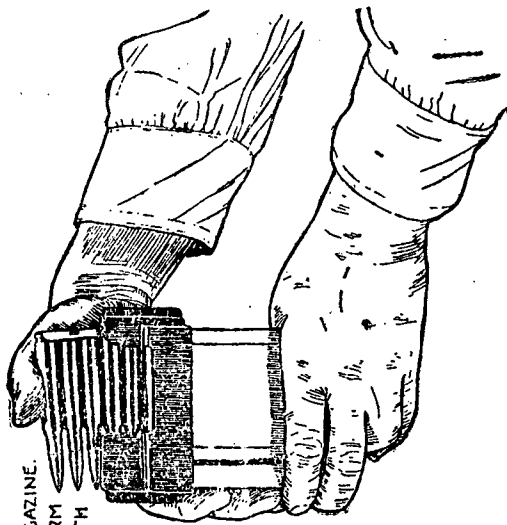
■ 32. TO LOAD THE RIFLE.—Press the magazine release, withdraw the empty magazine, if it has not already fallen out. Hold a loaded magazine with its base in the palm of the right hand, cartridges pointing to the front. Insert the magazine between the magazine guides in front of the trigger guard and push it home. Tap the bottom of the magazine with the right hand to insure that it is properly seated. The magazine can be inserted with the operating parts in either the rearward or forward position. It is ordinarily inserted after the rifle has been cocked.

■ 33. TO UNLOAD THE RIFLE.—Press the magazine release and withdraw the magazine, if it has not already fallen out. Let the bolt go forward by pulling the trigger.

■ 34. TO SET THE CHANGE LEVER.—*a.* For the slow cyclic rate of automatic fire, push the change lever to the forward position marked "F."

b. For the normal cyclic rate of automatic fire, set the change lever in the vertical position against the change lever stop marked "A."

c. To set the rifle at safe, depress the change lever stop and pull the change lever rearward until it covers the change lever stop. This position is marked "S." The change lever



1. PLACE FILLER OVER MAGAZINE.
2. REST MAGAZINE ON FIRM BASE AND STEADY WITH LEFT HAND.
3. SEAT CLIP IN FILLER AND PUSH DOWN WITH THUMB AS IF LOADING RIFLE.

FIGURE 14.—To load the magazine.

stop prevents the accidental setting of the change lever at safe and, at the same time, permits a quick change from safe to either normal cyclic rate or slow cyclic rate.

■ 35. TO FIRE THE RIFLE.—Select the cyclic rate desired and press the trigger, holding it for the length of burst desired.

If single shots are desired, set the change lever for the slow cyclic rate, press the trigger, and immediately release it as the slide begins its forward movement.

■ 36. GAS ADJUSTMENT.—*a. General.*—(1) The rifle normally should be operated on the smallest port, and this setting will never be varied unless the rifle shows signs of insufficient gas. To aline the smallest port, screw the gas cylinder all the way into the gas cylinder tube, turn it back one complete turn, and then, by turning the gas cylinder through the shortest arc, place the smallest circle on the cylinder head toward the barrel. Lock the cylinder in position. If, upon firing, the rifle shows signs of insufficient gas, try to secure a better alinement of the *small port*, either by backing the cylinder off one complete turn or by screwing it in one complete turn. As soon as the proper setting has been obtained, the rifleman must note the position so that he can quickly assemble the cylinder to the proper point without trial. If the rifle still shows signs of insufficient gas, then set the cylinder on the next larger port by the method described above for the initial setting.

(2) The larger ports are provided for use in case the action of the rifle has been made sluggish through the collection of dirt and grit or the lack of oil under conditions which render prompt correction impracticable. For this reason the threads should be kept clean and oiled and the cylinder free to turn. The extractor, ejector, and the chamber of the barrel should be examined and cleaned and defects corrected when possible. Under adverse conditions, and when signs of insufficient gas become apparent, the cylinder should be unscrewed one-third turn, thus registering the medium circle and alining the medium port with the gas orifice. Repeat this operation in order to connect the largest port with the barrel port.

b. Results of insufficient gas.—(1) Failure to recoil (usually due to misalined or excessively clogged gas port or extremely dirty mechanism).

(2) Failure to eject.

(3) Weak ejection.

(4) Uncontrolled automatic fire (exceptional).

c. *Results of too much gas.*—(1) Excessive speed causing pounding.

(2) Excessive heat in gas operating mechanism.

■ 37. **SAFETY PRECAUTIONS.**—For safety precautions to be exercised in handling and firing the automatic rifle see paragraph 241.

SECTION VI

STOPPAGES AND IMMEDIATE ACTION

■ 38. **GENERAL.**—A stoppage is any unintentional cessation of fire. For combat efficiency, the automatic rifleman must understand the *cause* of common stoppages, observe a definite routine to *prevent* the cause, and act promptly to *reduce* a stoppage should one occur.

■ 39. **CAUSES OF STOPPAGES.**—The following table gives the usual cause(s) of common stoppages:

Nature of stoppage	Usual causes	Other causes
Failure to feed.....	Dirty or dented magazine; weak magazine spring; worn notch on magazine rib or spread lips at magazine opening.	Empty or ruptured cartridge case in chamber; excessive friction in operating parts.
Insufficient gas.....	Dirty gas cylinder; clogged gas ports; improper adjustment of gas cylinder; gas cylinder not oiled.	
Sluggish operation of parts.	Dirty parts; lack of oil; weak recoil spring; oil too heavy for cool weather.	Excessive oil (especially in chamber) may collect dirt or become sticky after heating.
Defective or improperly assembled parts.	Worn or broken firing pin, extractor, magazine catch, sear spring.	Improper or incomplete assembly such as incorrectly placing, or omitting, the connector.
Faulty ammunition.	Defective primer; lack of or reduced powder charge; bulged round.	

■ 40. **PREVENTION OF STOPPAGES.**—Almost all stoppages can be prevented by timely care and cleaning of the automatic rifle as set forth in paragraph 14 and in the following table:

POINTS TO BE OBSERVED BEFORE, DURING, AND AFTER FIRING

Item	Before	During (including temporary cessations)	After. (See par. 14.)
Bore.....	Remove oil.....	Keep free from dirt, mud, water and snow; clean during pauses.	Clean and oil.
Chamber.....	Clean thoroughly with chamber cleaning brush and patches.	Clean frequently. Use chamber cleaning brush during pauses while rifle is hot. Keep free from oil.	Clean thoroughly. Oil lightly.
Operating parts.	Work slide back and forth to insure free movement. Oil lightly if necessary.	Oil periodically. Keep out dirt and dust.	Clean and oil.
Gas system....	Verify proper setting of gas port.	Place 3 or 4 drops of oil in end of gas cylinder periodically during pauses.	Clean and oil.
Trigger mechanism.	Test at safe, slow and normal cyclic rate settings.	-----	Clean and oil.
Magazines....	Clean; remove dents; see that springs and catch function properly.	Handle carefully; keep out dirt and grit; avoid throwing in sand.	Disassemble; clean and oil.

■ 41. IMMEDIATE ACTION.—*a.* Immediate action is the procedure used to reduce common stoppages in the minimum time. This procedure, given in the table below, consists of a positive and unhesitating manual operation without detailed consideration of the cause.

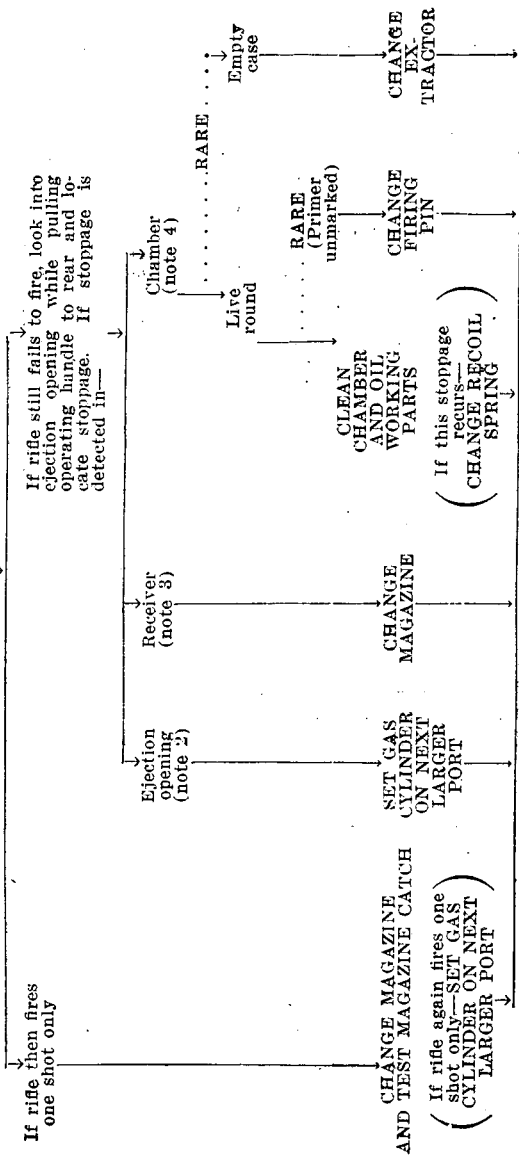
b. It will be seen that the first phase (Pull, Push, Tap, Aim, Fire) is an automatic action, and that the second phase, when necessary, consists of only enough analysis and inspection to provide quick and positive action.

c. Ruptured cartridges may result from any one of several causes—excessive head space, dirty or sticky chamber, or defective cartridge cases.

d. If the rifle on a cold day fires single shots when the change lever is set on slow cyclic rate, it is due to sluggishness in the rate reducing mechanism or too heavy oil on the

RIFLE FAILS TO FIRE IMMEDIATE ACTION

(Pull, Push, Tap, Aim, and Fire) (note 1)



operating parts. Set the change lever on the normal cyclic rate until the rifle warms up.

e. In the event of stoppages that can not be remedied, the rifle should be reported for repair. (Such a stoppage would be the repeated rupturing of cartridges near the base due to excessive head space.)

NOTES

1. *Pull* operating handle to rear; *Push* operating handle forward; *Tap* up on magazine; *Aim*, and attempt to *Fire*. *This action will relieve 90 percent of all stoppages.*

2. This stoppage will be detected by empty case wedged in ejection opening by live round attempting to get into chamber.

3. This stoppage will be detected by absence of cartridge in receiver and chamber, or two cartridges attempting to get into chamber at same time, or bolt riding over cartridge and wedging it while it is halfway in chamber.

4. This stoppage will be detected by live round jammed into ruptured cartridge in chamber, or live round in chamber with primer marked, or (RARE) live round in chamber with primer unmarked, or (RARE) empty case in chamber.

SECTION VII

SPARE PARTS AND ACCESSORIES

■ 42. SPARE PARTS (fig. 15).—a. The parts of any rifle in time will become unserviceable through breakage or wear resulting from continuous usage. For this reason, spare parts are provided for replacement of the parts most likely to fail. These parts are for use in making minor repairs and in the general upkeep of the rifle. Twenty-round magazines also are issued as spares, the quantity being based on the allowance of ammunition authorized. Sets of spare parts should be maintained as complete as possible at all times, and all parts should be kept clean and lightly oiled to prevent rust. Whenever a spare part is used to replace a defective part in the rifle, the defective part should be repaired and substituted in the spare parts set, or a new one should be obtained. Parts that are carried in complete sets at all times should be assembled correctly and ready for immediate insertion in the rifle. The allowances of spare parts and of 20-round magazines are prescribed in Standard Nomenclature List No. A-4.

b. With the exception of the spare parts mentioned in

paragraph 43b (4); repairs or alterations by the using organizations are prohibited.

■ 43. ACCESSORIES.—*a. General.*—Accessories include the tools required for disassembling and assembling and for the cleaning and preservation of the rifle. They must not be used for any purpose other than as prescribed. There are a number of accessories the names or general characteristics of which indicate their uses or application. Therefore, detailed description or methods of use of such items are not outlined herein. However, accessories embodying special features or having special uses are described in *b* below.

b. Special.—(1) *Brush and thong, caliber .30, complete.*—This consists of the brush, the tip, the weight, and the cord. The thong weight and tip are made of brass and are provided with holes in which the thong cord is knotted. The tip is provided with a cleaning patch and is threaded on the end to receive the fiber bristle brush.

(2) *Brush, cleaning, caliber .30, M2.*—The brush consists of a brass wire core with brass bristles and tip. The core is twisted in a spiral and holds the bristles in place. This brush is distinguished from the one that is a part of the brush and thong by the type of bristles and lengths of threads on the tip, and is for use with the cleaning rod. It can be used with the thong only with difficulty, because of its stiff bristles.

(3) *Brush, chamber-cleaning, M1.*—The chamber-cleaning brush consists of a curved, flat, steel handle to which are hinged a chamber-cleaning brush at one end and a small, bristled dusting brush at the other end.

(4) *Case, accessory, and spare parts, M1918.*—This is a leather box-shaped case, approximately 2 inches wide, 4 inches high, and 6 inches long. This case should be carried at all times, and will normally be complete with the following spare parts and small accessories:

(a) Spare parts.—

1. Recoil spring.
2. Connector.
3. Two trigger pins.
4. Two firing pins.

5. Extractor and extractor spring.
6. Trigger guard retaining pin and spring.
7. Gas cylinder tube retaining pin and spring.
8. Sear spring.
9. Change lever spring.
10. Magazine catch spring.
11. Change lever stop and spring.
12. Change lever.

(b) *Accessories.*—

1. Fabric envelope, 3 by 5 inches, two-button.
2. Fabric envelope, 3 by 3 inches, one-button.
3. Oval oiler, 3-ounce with cap and chain (full of oil).
4. Chamber cleaning brush M1.
5. Combination tool.
6. Gas cylinder cleaning tool.
7. Ruptured cartridge extractor.
8. Brush and thong, complete.

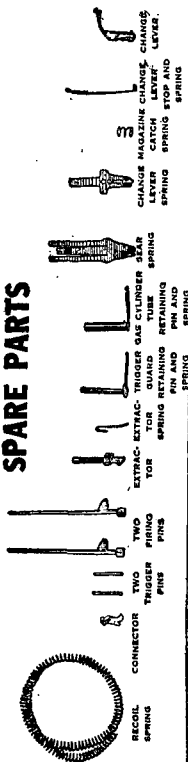
(5) *Case, cleaning rod M1.*—This fabric case has five pockets, four of which hold the sections of the cleaning rod M1, while the fifth holds the cleaning brush, caliber .30, M2. The contents are secured in their pockets by a web flap and a buckle.

(6) *Rod, cleaning, jointed, M1.*—This steel rod consists of five sections, the first two permanently fastened together by a swivel joint. The first section has a slot for attaching the cleaning patch, and a threaded hole is provided for holding the cleaning brush. The rear section is provided with a tubular steel handle which turns on the rod. There are many other types of cleaning rods in use in the military service that may be used to clean the automatic rifle in garrison or bivouac. The cleaning rod M1, or the brush and thong complete, are the most practical cleaning accessories in the field or in combat.

(7) *Filler, magazine.*—The magazine filler is a pressed-steel adapter which is fitted over the top of an empty magazine when loading. Its method of use is shown in figure 14.

(8) *Extractor, ruptured cartridge, Mk. II.*—The ruptured cartridge extractor has the general form of a caliber .30 cartridge. It consists of three parts—the spindle, head, and sleeve. To use the ruptured cartridge extractor, the live car-

SPARE PARTS



ACCESSORIES

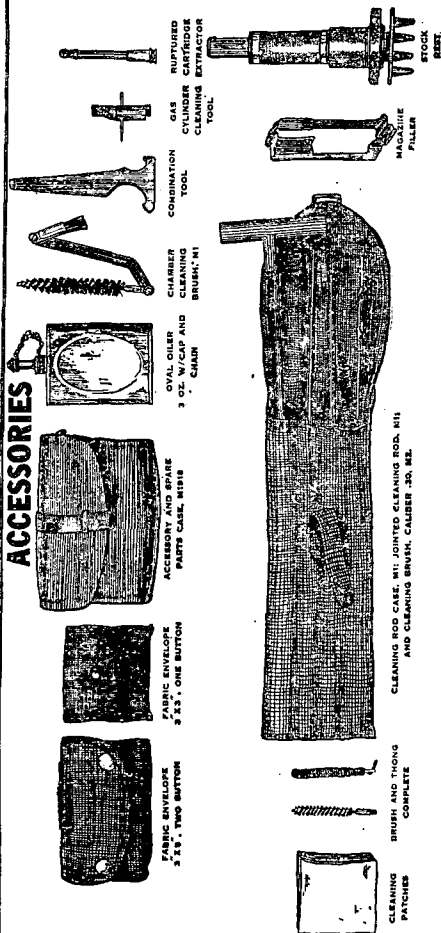


Figure 15—Spare parts and accessories.

tridges must be removed from the rifle. The ruptured cartridge extractor is then inserted through the opening of the ruptured cartridge case and pushed forward into the chamber. The bolt is let forward without shock so that the extractor of the rifle engages the ruptured cartridge extractor. As the operating handle is drawn back, the ruptured cartridge extractor, holding the ruptured cartridge on its sleeve, is extracted.

(9) *Sling, gun, M1907.*—The gun sling is fastened to the swivels provided on the rifle and is always carried thereon. It consists of a long and a short strap, either of which may be lengthened or shortened to suit the particular soldier using it.

(10) *Tool, combination.*—This tool consists of a steel body having two spanner wrenches and a screw driver end. The small spanner is used to turn the gas cylinder, and the large spanner will fit the rear of the barrel. The large screw driver is used to remove or tighten the forearm screws.

(11) *Tool, cleaning, gas cylinder.*—This is a special tool for cleaning the gas-operating mechanism. The ends of the tool body may be used to scrape carbon from the interior of the gas cylinder and from the face of the gas piston. The drift, which is attached to the body, may be used to remove carbon deposits from the gas ports and the grooves of the gas piston. The carbon must be completely removed, but care must be exercised to avoid scoring or damaging the gas cylinder walls or the grooves of the gas piston.

(12) *Wrench.*—There is no special wrench issued to remove the flash hider and bipod bearing from the muzzle end of the rifle barrel. Any suitable wrench that will fit a 1-inch nut may be used for this purpose.

(13) *Stock rest.*—The stock rest is used when it is desired to employ the automatic rifle for the delivery of fire on a final protective line, or to fire on a fixed point when the visibility is poor (par. 239).

SECTION VIII

AMMUNITION

■ 44. *GENERAL.*—Information in this section describes the several types of cartridges authorized for use in the Browning

automatic rifle, caliber .30, M1918A2, means of identification, care, use, and ballistic data.

■ 45. **CLASSIFICATION** (fig. 16).—Based upon use, the principal classifications of ammunition for this rifle are—

- a. *Incendiary*, for use against inflammable matériel.
- b. *Armor piercing*, for use against lightly armored vehicles, concrete shelters, and personnel.
- c. *Tracer*, for observation of fire and incendiary purposes.
- d. *Ball*, for use against personnel and light matériel targets.
- e. *Blank*, for simulated fire.
- f. *Dummy*, for training. (Cartridges are inert.)

■ 46. **LOT NUMBER**.—When ammunition is manufactured, an ammunition lot number, which becomes an essential part of the marking, is assigned in accordance with specifications. This lot number is marked on all packing containers and on the identification card inclosed in each packing box. It is required for all purposes of record, including grading and use, reports on condition, functioning, and accidents in which the ammunition might be involved. Since it is impracticable to mark the ammunition lot number on each individual cartridge, every effort should be made to maintain the ammunition lot number with the cartridges after they are removed from their original packing. Cartridges which have been removed from the original packing and for which the ammunition lot number has been lost are placed in grade 3. It is therefore necessary when cartridges are removed from original packings that they be so marked that the ammunition lot number is preserved.

■ 47. **GRADE**.—Current grades of existing lots of small-arms ammunition are established by the Chief of Ordnance and are published in Ordnance Field Service Bulletin No. 3-5.

■ 48. **IDENTIFICATION**.—a. *Markings*.—The contents of original boxes are readily identified by the markings on the box. Similar markings on the carton label identify the contents of each carton.

b. *Color bands*.—Color bands painted on the sides and ends of the packing boxes further identify the various types

of ammunition. The following color bands for cartridges are used:

Incendiary-----	red on yellow
Armor piercing-----	blue on yellow
Tracer-----	green on yellow
Ball-----	red
Blank-----	blue
Dummy-----	green

c. *Types and models.*—The following types and models of caliber .30 cartridges are authorized for use in this rifle:

Incendiary.
 Armor-piercing, M2.
 Tracer, M1.
 Ball, M2.
 Ball, M1.
 Blank, M1909.
 Dummy, M1906 (corrugated).

When removed from their original packing container, the cartridges are identified, except as to ammunition lot number and grade, by their physical characteristics.

(1) *Incendiary.*—All caliber .30 incendiary ammunition is distinguished by the tip of the bullet nose, which is painted light blue. The bullets have gilding metal jackets.

(2) *Armor piercing.*—All caliber .30 armor-piercing ammunition is distinguished by the tip of the bullet nose, which is painted black. The bullets have gilding metal jackets.

(3) *Tracer.*—Caliber .30 tracer ammunition may be identified by the tip of the bullet nose, which is painted red.

(4) *Ball.*—All models of caliber .30 ball ammunition which are authorized to be used in the automatic rifle, except M2 of 1940 and earlier manufacture, have bullets with gilding metal jackets. The metal jacket of the earlier M2 ammunition is tin-coated. The gilding metal jacket of the present M2 bullet is copper-colored.

(5) *Blank.*—Blank ammunition may be identified by the absence of a bullet in the cartridge case.

(6) *Dummy*.—The caliber .30 corrugated dummy cartridge may be identified by the corrugations formed in the cartridge case.

■ 49. CARE, HANDLING, AND PRESERVATION.—*a*. Small-arms ammunition is not dangerous to handle. Care, however, must be exercised to keep the boxes from becoming broken or damaged. All broken boxes must be repaired immediately and all original markings transferred to the new parts of the box. The metal liner should be air-tested and sealed, if equipment for this work is available.

b. Ammunition boxes should not be opened until the ammunition is to be used. Ammunition removed from the airtight container, particularly in damp climates, is likely to corrode, thereby causing the ammunition to become unserviceable.

c. The ammunition should be protected from mud, sand, dirt, and water. If it gets wet or dirty, wipe it off at once. Light corrosion, if it forms on cartridges, should be wiped off. However, cartridges should not be polished to make them look cleaner or brighter.

d. During marksmanship and combat training, no caliber .30 ammunition will be fired until it has been identified by ammunition lot number and grade.

e. Do not allow the ammunition to be exposed to the direct rays of the sun for any length of time. This is likely to affect seriously its firing qualities. Small-arms ammunition of all classes should be stored away from radiators, hot water pipes, and other sources of heat, since heat not only deteriorates ammunition, but also creates a fire hazard.

■ 50. STORAGE.—Whenever practicable, small-arms ammunition should be stored under cover. Should it be necessary to leave small-arms ammunition in the open, it should be raised at least 6 inches from the ground and covered with a double thickness of tarpaulin. Suitable trenches should be dug to prevent water from flowing under the pile.

■ 51. BALLISTIC DATA.—The approximate maximum ranges of the several types of models of caliber .30 ammunition

authorized for use in the Browning automatic rifle are shown below:

	Yards
Incendiary (Incendiary action up to 400 yards)-----	
Armor piercing, M2-----	4,000
Tracer, M1-----	3,500
Ball, M2-----	4,000
Ball, M1-----	5,500
Blank-----	20
Dummy, MI906 (corrugated)-----	None

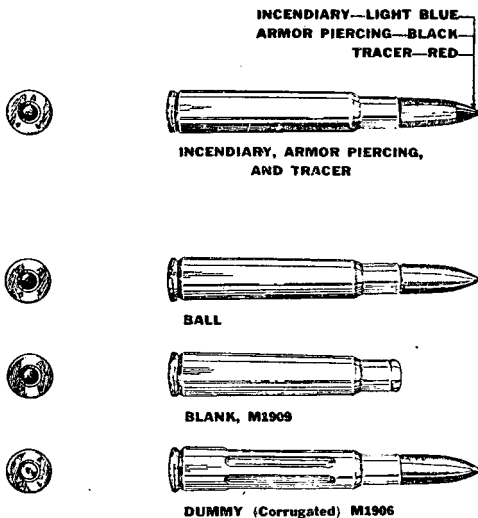


FIGURE 16.—Ammunition.

CHAPTER 2

MARKSMANSHIP, KNOWN DISTANCE TARGETS

	Paragraphs
SECTION I. General.....	52-54
II. Preparatory marksmanship training.....	55-80
III. Courses to be fired.....	81-83
IV. Range practice.....	84-93
V. Regulations governing record practice.....	94-124
VI. Targets and ranges.....	125-126

SECTION I

GENERAL

■ 52. **OBJECT.**—The object of this chapter is to provide a thorough and uniform method of training individuals to be good automatic rifle shots, and of testing their proficiency in firing at known distance targets with the Browning automatic rifle, caliber .30, M1918A2. This proficiency is vital to the effective use of the weapon in field firing, both for aimed fire and other types of fire (such as fire along a final protective line, par. 239), described in succeeding chapters.

■ 53. **FUNDAMENTALS.**—To become a good automatic rifle shot, the soldier must be trained in the following essentials of good shooting:

- a. Correct sighting and aiming.
- b. Correct positions.
- c. Correct trigger manipulation.
- d. Correct application of the fundamentals of automatic fire.
- e. Knowledge of proper sight adjustments.

■ 54. **PHASES OF TRAINING.**—a. Marksmanship training is divided into the following phases:

- (1) Preparatory marksmanship training.
 - (2) Range practice.
- b. No individual should be given range practice until he has had a thorough course in preparatory training.